

- 1(a) A team of programmers create a robot that will be used in a factory. The robot will be able to do the work of multiple humans.

The programmers discuss whether to write the instructions for the robot in assembly language or a high-level language.

The robot provides a web-based interface for users. The home screen webpage for this interface is shown in Fig. 4.

Robot User Interface

Robot prime directives

- Serve the company trust
- Protect data
- Uphold standards

[Updates](#)

Login

Password

Fig. 4

- i. Complete this HTML code that will display the webpage shown in Fig. 4.

```
<html>
  <head>
    <title>Robot User Interface</title>
  </head>
  <body>
    <h1>Robot prime directives</h1>
    .....
    <li>Serve the company trust</li>
    <li>Protect data</li>
    <li>Uphold standards</li>
    .....
    <a ..... = "updates.html">Updates</a>
    <p>.....</p>
    <form action="dologin.php">
      Password
      <input type = "....." name="pw">
      <input type = ".....">
```

```
</form>
</body>
</html>
```

[5]

- ii. Write CSS code that could be used in an external stylesheet to format all text using the <h1> tag as white with a red background.

[3]

(b) The robot’s web interface uses images that show the robot in action. These photographs have been taken using a digital camera.

The programmers do not want other people to download and use these images.

i. State the name of **one** relevant piece of legislation and describe how this would protect these images.

Legislation -----

Description -----

[3]

ii. For other areas of the web interface, the programmers need to use images that they have not created themselves.

Give **two** ways that they could make sure these images are used legally.

1 -----

2 -----

[2]

2 The following JavaScript has been found to crash certain web browsers.

Line	Code
1	var total = "";
2	for(var j = 0; j < 200000; j++)
3	{
4	total = total + j.toString();
5	history.pushState(0,0, total);
6	}

j.toString() converts j to a string. It is the JavaScript equivalent to str(j).

Complete the table below.

Line	Effect of Code
1	
2	
3	
4	
5	Pushes total onto a stack that holds the browser's history.
6	

[2]

[1]

[1]

- 3 A website sells tickets for sporting events. The website uses HTML, CSS and JavaScript.

Describe the purpose of HTML and CSS within the code of the website.

HTML

CSS

[4]

- 4 Shreya is a web developer who creates webpages for a variety of different companies.

Shreya has received a webpage written in HTML shown in Fig. 2.1.

```
<html>
  <head>
    <title>BBC Computer</title>
    <link rel="stylesheet" type="text/css" href="BBC.css">
  </head>
  <body>
    <h1 id="Beeb">History of the BBC Micro Computer</h1>

    <h2>BBC Micro Models</h2>
    <ul>
      <li class="model">Model A</li>
      <li class="model">Model B</li>
    </ul>

  </body>
</html>
```

Fig. 2.1

Some of the formatting requirements for this webpage are shown in Fig. 2.2.

		Formatting Requirements
--	--	-------------------------

- 5 A web development company makes its money building websites for other companies.

The site also contains the following code.

```
<div class="offer">All oranges 50% off.</div>
```

Complete the CSS code that would make any div elements of the class offer have an orange border.

```
.....{  
  
        border-style: solid;  
  
        .....  
    }
```

[2]

- 6 One page in a website contains a hyperlink on an image. When the image stored as “ticket.png” is clicked, the user is hyperlinked to the page stored as “booking.htm”.

Write the HTML code to implement this hyperlink.

```
-----  
  
-----  
  
-----
```

[3]

Part of the site is shown below. The words 'Book tickets' link to the page 'bookings.html'.

1. Macbeth
2. Blood Brothers
3. An Inspector Calls

Book tickets

Write the HTML code for the extract above.

[3]

-----[3]

iii. Explain why using a RISC processor rather than a CISC processor is likely to result in increased battery life.

-----[3]

9(a) A website contains the following HTML:

```
<html>
<head>
  <title>Boris' Cake Shop</title>
  <link rel="stylesheet" type="text/css" href="style.css">
</head>
<body>
  <h1>Boris' Cake Shop</h1>
  <p id="intro">
    Welcome to Boris' cake shop.
    <script>
      var hour = new Date().getHours();//gets the hour value of the
current time
      if(hour>9 && hour<17)
      {
        document.write("We are currently open.");
      }
      else
      {
        document.write("We are closed, come visit us when we are
open (09:00 - 17:00).");
      }
    </script>
  </p>
  <div class="customerQuote">
    Boris makes the best cupcakes I have ever tasted.
  </div>
</body>
</html>
```

Explain the difference between a HTML id attribute and a HTML class attribute.

----- [2]

(c) The code between the `script` tags is supposed to display a different message during the shop's opening hours of 09:00 and 17:00.

i. State the name of the language used between the `script` tags.

-----[1]

Some users have reported that there is a bug and the site says the shop is closed when they log on between 9 and 10 in the morning.

ii. Explain how this bug can be fixed.

-----[1]

iii. Give **one** disadvantage of this code being run client side rather than server side.

-----[1]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1	a	i	• and • href • Login • text/password • submit	5	<pre> <html> <head> <title>Robot User Interface</title> </head> <body> <h1>Robot directives</h1> Serve the company trust Protect data Uphold standards Updates <p>Login</p> <form action="dologin.php"> Password <input type = "text" name="pw"> <input type = "submit"> </form> </body> </html> </pre> HTML tags are not case sensitive Correct answer only <u>Examiner's Comments</u> Generally well answered and many candidates gained full marks with most being able to gain at least two.
		ii	• h1 and other code contained in { } • color :white; • background-color : red; //background: red;	3	Ignore presence or lack of <style> tags. Ignore lack of semicolons Penalise misspelling of "color" once and then FT <pre> h1 { color:white; background-color: red; } </pre> White can be #FFFFFF or #FFF Red can be #FF0000 or #F00 <u>Examiner's Comments</u> Many candidates were able to gain full marks. The most common reasons candidates lost marks was through the use of quotes around the colour equals instead of colons and for misspelling colour

Question			Answer/Indicative content	Marks	Guidance
	b	i	• <u>Copyright Designs and Patents Act</u> Any two from: • Gives the author (the programmers) ownership/copyright of the photographs • ...no need to apply // this is automatic • Others cannot use/distribute // can be prosecuted/fined for using/distributing... • ...without permission • Permission can be granted / bought / licenced	3	Must be full name of Act for MP1 FT for versions of Copyright or nothing for MP2-6 <u>Examiner's Comments</u> Many candidates were able to gain 2 marks but many did not give the full name of the legislation.
		ii	• Ask permission of author / photographer / owner • Use images marked as copyright free (e.g. Creative Commons Licence) • Purchase (licence to use) image	2	Do not accept just "ask permission" <u>Examiner's Comments</u> Many candidates were able to gain 2 marks. It was surprising to see how many believed that you could use copyright images for your business just by crediting the artist. Candidates should be made aware that although crediting the artist may help avoid plagiarism it does not allow you free use of a copyright image.
			Total	13	
2			1. Creates / declares / defines a variable (called total) (1) and assigns it an empty / blank (string). (1) 2. A loop that iterates 200,000 times. (1) 3. - 4. Concatenates (the string version of) j to total. (1)	4 AO3.3	For point 1 accept 'blank value' for empty string For point 4 accept <i>add</i> instead of <i>concatenates</i> <u>only</u> if it is clear it is building a string and not adding a numeric value. Accept 'append'. <u>Examiner's Comments</u> In general, candidates did not use technical terminology when describing the effect of the code e.g. declaring a variable, assigning an empty string, concatenating strings. Centres should encourage candidates to use these and similar terms in response to questions where they are required to describe code.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
3			• HTML defines the structure of a web page • HTML defines the content of a web page • Using tags (enclosed in <>) • CSS defines the style / appearance • Using selectors such as classes / IDs / etc • Can be placed within HTML or externally in a file • Multiple pieces of CSS can be combined (the more local instances overriding)	4 AO1.1	Do not accept layout/format for HTML as this is too vague and can be used to describe CSS
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
4			1 mark per bullet up to a maximum of 6 marks: • Correct syntax for css class for “model”(including opening and closing curly brackets) • Correct syntax for css id for “beeb” (including opening and closing curly brackets) • Correct syntax for setting the font family to arial for class “model” • Correct syntax for setting the font colour to red on class “model” • Correct syntax for setting colour to green for id “beeb” • Correct syntax for setting font family to times for id “beeb”	6 AO3.1 (6)	Example: <pre>li.model { color: red; font-family: arial; } #Beeb { color: green; font-family: times; }</pre> Ignore if li is not put in front of .model Reject if any other tag is put in front of .model Responses written in HTML mark as NAQ
			Total	6	
5			<pre>.offer{ border-style: solid; border-color: orange; }</pre>	2 AO3.2	Accept div. offer Accept hex / RGB codes that would provide a shade of orange. Closing ; is optional <u>Examiner's Comments</u> It was evident that some candidates did not have the required awareness of the CSS class and property definitions outlined in the specification appendix 5d. Those candidates who did, generally scored well on this question.
			Total	2	
6			• Correct <a> with close • href property to correct page • correct tag with src property to correct file	3 AO3.2	<pre></pre>
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
7			– <code></code> for ordered list – <code></code> for each item – <code></code> and <code></code> around Book tickets For points 1 and 2 also allow numbers typed in providing line breaks have been added. e.g. 1. Macbeth
 2. Blood Brothers
 3. An Inspector Calls
	3 (AO3.2)	Upcoming productions: <pre> Macbeth Blood Brothers An Inspector Calls Book tickets</pre> Examiner's Comments It was evident that some candidates did not have the required awareness of the HTML tags outlined in the specification appendix. Those candidates who did, generally scored well on this question. See Exemplar 1 which was given full marks. Exemplar 1 <pre><p> * Upcoming productions: </p> Macbeth Blood Brothers An Inspector Calls
 Book tickets</pre>
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
8		i	- Tags to make "Features" a heading (accept h1, h2, h3 etc.) - Correct use of ol - Correct use of li tags - Use of <a tag Around the text "Download the Factsheet" - correct use of href="factsheet.pdf" (1 Mark per -, max 5)	5 (AO3.2)	For making Features a heading only accept strong/b if accompanied by code to increase font size. <pre><h1>Features</h1> The new OCR Smart Watch: Uses the CB2 RISC processor for long battery life Stores up to 20hrs of music Tracks fitness Download The Factsheet</pre> Li close tags are optional <u>Examiner's Comments</u> Surprisingly fewer candidates than anticipated gained full marks on this question. Many candidates gained some marks. Marks were invariably lost on the HTML for the hyperlink.
		ii	- A program called a spider/crawler/bot - Traverses the web / following the links. - It takes each word in the document - ...It adds an entry for the page (under the word) in the index... - ...alongside the word's position on the page. (1 Mark per -, Max 3)	3 (AO2.1)	<u>Examiner's Comments</u> This question was poorly attempted. Many candidates mentioned a crawler/spider program but then failed to clearly describe the process. Some candidates went on to discuss ranking even though the question specifically stated not to.
		iii	- RISC has a smaller instruction set (than CISC) - Requires fewer transistors / less complex circuitry - Means less power is required. (1 Mark per -, Max 3)	3 (AO1.2)	<u>Examiner's Comments</u> Most candidates achieved two marks on this question with few referring to the fact that RISC requires less complex circuitry.
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
9	a		Only one element can have a given id / id is unique. (1) Class can be used assigned to multiple elements / used multiple times. (1)	2 (AO1.1)	Examiner's Comments Those candidates who achieved credit on this question, generally achieved both marks.
	b		<pre> hl{ (1 mark for open and close) font-family:Arial(;) (1 mark) } .customerQuote{ (1 mark) background-color: #E8C3E1(;) (1 mark) } #intro{ (1 mark) (font-)color: darkRed(;) (1 mark) } </pre>	6 (AO3.1)	.customerQuote must have . and opening and closing { } for 3rd mark. #intro must have # and opening and closing { } for 4th mark Must match case sensitivity, except for 'Arial' and 'darkRed' and colour code Allow quotes around Arial and darkRed Examiner's Comments Few candidates scored more than two marks on this question. There was a general lack of attention to detail resulting in fundamental mistakes e.g. missing close bracket }; equals (=) instead of colon (:) when setting attributes.
	c	i	JavaScript	1 (AO1.1)	Cao do not accept Java Examiner's Comments Well attempted by most candidates with many scoring two out of a possible three marks. Some candidates did not gain credit in iii) because they cited '<i>...time taken to process</i>' as a disadvantage.
		ii	Change line if(hour>9 && hour<17) To if(hour>8 && hour<17) (1 Mark) or ... if(hour>=9 && hour<17) (1 Mark)	1 (AO3.3)	Accept Change 'greater than' to 'great than or equal to' or similar

Question			Answer/Indicative content	Marks	Guidance
		iii	– Won't work if JavaScript is disabled. (1) – Shows incorrect message if user's computer's clock is wrong / in different time zone. (1) – (Source) code is visible allowing it to be copied / modified. (1) (Max 1)	1 (AO2.2)	
			Total	11	